

🛡 SpoofVane — Complete Feature Catalogue

Everything in the product: 76 backend capability modules, 7 Bright Data products, the platform & governance layer, and 21 console screens.

v0.5.0 · built by Claude (Opus 4.8) in Claude Code · 29 May 2026, 09:57 BST

76

capability
modules

7

Bright Data
products

21

console screens

350

tests passing

15k+

lines of code

Capability modules — 8 groups, 76 features

● Discovery 10 modules

How SpoofVane finds impersonation

A1

SERP scanner

Bright Data SERP API
(Google/Bing/Brave/DuckDuckGo, brd_json=1);
paginates & dedupes.

A2

Cert transparency

Tails CT logs (Calidog certstream) + Web Scraper
API fallback for new lookalike certs.

A3

DNS brute

Permutation + bitsquatting + homoglyph + IDN;
resolves via residential proxies.

A4

Registrar feed

Bright Data Datasets WHOIS feed; flags newly-
registered & risky-country domains.

A5

Phishing feeds

OpenPhish + PhishTank + URLhaus + UrlAbuse
ingest, normalised to one IOC schema.

A6

Social impersonation

X / IG / TikTok / LinkedIn / YouTube via SERP +
Web Unlocker.

A7

A8

URL shortener

Expands bit.ly/t.co/lnkd.in chains, recording every redirect hop to the terminal URL.

Ad network

Google/Meta/TikTok ad libraries via Web Scraper API; captures creative+advertiser+landing.

A9

Mobile app store

iOS App Store + Google Play scraping for fake apps; matches name + icon pHash.

A10

Crawl seed

Customer seed lists + dark-web mention adapters (HudsonRock/SpyCloud pattern), idempotent.

● Inspection 8 modules

How each suspect is captured & analysed

B1

Browser inspector

Bright Data Scraping Browser over CDP, country-pinned; DOM+screenshot+HAR+final URL.

B2

Cloaking detector

Renders ≥ 2 geos and diffs DOM+pHash+content to catch geo-cloaking.

B3

DOM extractor

Classifies forms — login / payment / iframe — from the real parsed DOM.

B4

TLS inspector

Cert chain + JA3/JA4 fingerprint + SCT; flags self-signed.

B5

HAR collector

Full HAR with per-resource SHA-256, waterfall timing, external-origin tagging.

B6

pHash extractor

Perceptual + difference hashes on screenshots and logo crops.

B7

WHOIS enricher

WHOIS/RDAP + hosting ASN + rDNS; flags bulletproof hosts.

B8

Ad creative capture

Screenshot + OCR of an ad creative paired with its landing page.

● Scoring 10 modules

How similarity & risk are quantified

C1

C2

Composite scorer

Platt-calibrated probability (not a raw sum) with per-signal contribution breakdown.

URL risk scorer

Entropy + suspicious tokens + TLD + punycode + IP + age → risk with contributions.

C3

DOM scorer

Structural tree-edit distance vs known-good login pages.

C4

pHash scorer

Hamming distance over perceptual hashes vs brand assets.

C5

Logo scorer

CLIP/MobileCLIP logo detection (feature-flagged, never 0% from missing weights).

C6

Kit fingerprint

13 kit signatures (EvilProxy, Tycoon-2FA, Greatness, Mamba2FA...) via DOM + JS-bundle hash.

C7

Family classifier

ML classifier mapping kit features → threat family.

C8

Cluster score

Union-find threat-network clustering by shared ASN/registrar/kit/favicon/JA3.

C9

Voiceprint score

192-dim speaker embedding; cosine match against an enrolled executive voiceprint.

C10

Deepfake score

Ensemble: face embedding + lip-sync + C2PA provenance → deepfake probability.

● Verdict 8 modules

How the final call is made

D1

Claude verdict

Claude structured-output verdict with prompt-guard and recorded rationale.

D2

GPT verdict

GPT verdict for the ensemble, same schema.

D3

Gemini verdict

Gemini verdict for the ensemble, same schema.

D4

SLM triage

Small open-weight model — the cheap path for low-severity triage.

D5

Verdict merger

Disagreement-aware merge; discounts confidence and escalates to a human on dissent.

D6

Multimodal verdict

Fuses text + image + audio for deepfake verdicts.

D7

MITRE enricher

Maps every alert to real ATT&CK technique IDs + D3FEND countermeasures.

D8

Cache + active learning

Verdict cache keyed on evidence hash; analyst overrides retune calibration.

● Agents 10 modules

Autonomous, governed actions

E1

Takedown agent

Researches → drafts → HITL gate → submits registrar packets.

E2

Victim-id agent

Identifies victims from session telemetry (consent + lawful-basis gated).

E3

Cred-poison agent

Submits unique canary credentials per URL and tracks dark-web reuse.

E4

Synth-pages agent

Deploys honeypot pages back at the attacker in an isolated namespace.

E5

Cluster agent

Maintains the threat-network graph incrementally.

E6

Learning agent

Wires analyst overrides into the C1 calibration loop.

E7

Kill switch

Admin hard-halt for every running agent (global or per-tenant) + audit.

E8

Governance

Scores blast-radius; enforces budget/cost/risk envelope + HITL above threshold.

E9

Agent audit

E10

SLM triage agent

Hash-chained, tamper-evident entry for every agent action.

Agentic wrapper around the D4 cheap-path model.

● Delivery 8 modules

Takedown & integrations out

F1

Cloudflare takedown

Real Cloudflare abuse-form POST behind HITL; returns a reference id.

F2

GoDaddy takedown

GoDaddy abuse API submission.

F3

Namecheap takedown

Namecheap abuse API submission.

F4

Hosting abuse

AWS/GCP/Hetzner/OVH/Hostinger/Cloudflare per-host abuse channel routing.

F5

SIEM/SOAR sinks

Splunk + Sentinel + ServiceNow + PagerDuty + Cortex XSOAR, with retry/backoff.

F6

TAXII / STIX

Schema-valid STIX 2.1 bundle export over TAXII 2.1.

F7

Own MCP server

SpoofVane's MCP server — analyst-in-Claude-Desktop with 14+ tools.

F8

BD MCP client

Calls Bright Data's MCP server for ad-hoc BD-routed rendering.

● Platform 12 modules

Enterprise foundations

G1

Identity

OIDC + SAML + JWT + signed cookies + MFA + SCIM 2.0.

G2

RBAC

6 roles with explicit segregation of duties.

G3

Audit logger

G4

HITL review

Mandatory review queue; reviewer ≠ analyst.

SHA-256/Blake2b hash-chained audit, Ed25519-signed.

G5

Notifications

Email + Slack + Teams + WhatsApp + push + voice (Twilio/Plivo/Telnyx).

G6

Reports

Executive PDF + board pack + auditor pack, PKCS#7/PyHanko-signed.

G7

Rate limiter

Per-tenant token bucket.

G8

Idempotency

Idempotency-key middleware to make retried side-effects safe.

G9

Cost tracker

Per-tenant Bright Data + LLM spend telemetry vs envelope.

G10

Deepfake compliance

Chain-of-custody pack for legal discovery.

G11

BYOK

Customer-managed keys (AWS KMS / Azure KV / GCP KMS) via envelope encryption.

G12

Data residency

EU/US/APAC/IN routing + provable non-egress.

● AI Surfaces 10 modules

Analyst-facing intelligence

H1

Analyst copilot

Conversational verdict / triage assistant.

H2

Audit NL search

Natural-language search over the audit log → structured filters.

H3

Brand wizard

Conversational brand onboarding (keywords, TLDs, regions).

H4

Deepfake verdict UI

Analyst-facing deepfake verdict surface.

H5

H6

Exec attack surface

Executive likeness + voice surface monitor (sensitive, gated).

Family reranker

Re-ranks family classification with analyst feedback.

H7

Intel narrator

Auto-generates the human-readable alert narrative.

H8

Kit explainer

RAG over the phishing-kit knowledge base (prompt-cached).

H9

Takedown drafter

Drafts the registrar packet for analyst review.

H10

TTP proposer

Suggests MITRE TTPs for the alert.

Bright Data — 7 sponsor products

BD 1

Scraping Browser (CDP)

Country-pinned residential rendering of suspect pages

used by B1, B8, F1

BD 2

Residential Proxies

Geo-distributed HTTP + DNS from real residential exits

used by A3, A6, B1

BD 3

SERP API

brd_json=1 structured search across 4 engines

used by A1

BD 4

Web Unlocker

Bypasses JS challenges / 403s on protected targets

used by A6, A9, B1

BD 5

Web Scraper API

Structured extraction of ads & fallback pages

used by A2, A8, B8

BD 6

Datasets / Marketplace

WHOIS dataset feed for registration intelligence

used by A4, B7

BD 7

Bright Data MCP Server

Ad-hoc BD-routed rendering from inside the copilot

Platform & governance highlights



Replay mode

Every Bright Data path runs credential-free against input-dependent fixtures (reproducible build).



Differential depth probe

14 module probes assert output varies with input — stubs can't pass.



Two-person rule

Egress agents (cred-poison, synth-pages) need a second authoriser + permitted region.



Tamper-evident chains

Audit, agent-audit and chain-of-custody each hash-chain and self-verify.



Cost envelope

Per-tier caps (free \$5 / pro \$100 / business \$1k / enterprise \$25k) with SLM-first routing.



CycloneDX SBOM

26-component software bill of materials generated from the real dependency tree.

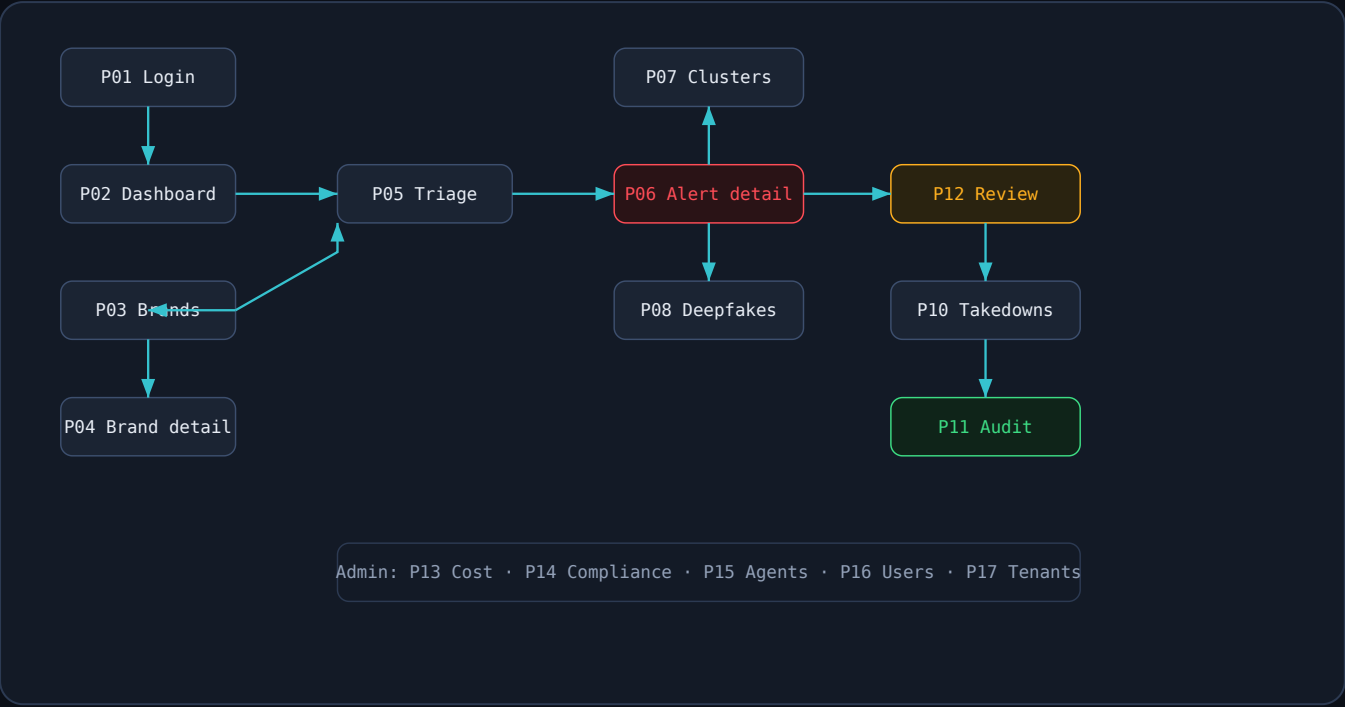
Modules map to the review's §8 catalogue (A1–H10) and the canonical page registry. Every feature has real, tested, input-dependent code – see `docs/TRACEABILITY_MASTER.md` and `docs/DEPTH_LEDGER.md`.

SpoofVane — Visual User Guide

Every screen, shown and annotated, plus how they connect.

v0.5.0 · built by Claude (Opus 4.8) in Claude Code · 29 May 2026, 09:53 BST · 21 screens (P01–P21)

Screen flow map



Cyan arrows show navigation. Red = the alert investigation hub; amber = human-in-the-loop review; green = the audit trail every action lands in. Admin screens sit in the bottom band.

The 21 screens

P01 Login



Continue with SSO (OIDC)

Continue with SAML

or

email

password

Sign in

🔒 MFA challenge appears here when enabled

Entry point. SSO/SAML/local + TOTP MFA. On success → Dashboard.

P02 Dashboard

app.spoofvane.io

SpoofVane

Dashboard

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Triage

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Threat posture

Active alerts

247

Phish today

18

Open
takedowns

9

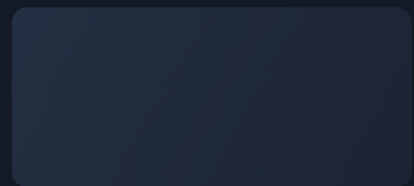
BD spend
(mo)

\$412

Threat families



Hosting regions



Live pipeline

discover → inspect → score → verdict → takedown

Day starts here. Click any KPI tile to drill into its queue.

P03 Brands

app.spoofvane.io

SpoofVane

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Protected brands

+ New brand (wizard)

Brand	Domain	Regions	Worst verdict
Acme Bank	acmebank.com	US, EU	BENIGN
Globex	globex.com	US	SUSPICIOUS
Initech	initech.io	APAC	PHISH

Brand list. The wizard (H3) suggests keywords, TLDs and regions for you.

P04 Brand detail

A screenshot of the SpoofVane application interface. The browser address bar shows 'app.spoofvane.io'. The application has a dark theme. On the left is a sidebar with the 'SpoofVane' logo (a hexagon icon) and two menu items: 'Dashboard' and 'Brands'. The 'Brands' item is highlighted with a dark blue background. The main content area has a large heading 'Acme Bank' in a white serif font. Below the heading are two rounded rectangular cards. The left card is titled 'Canonical assets' and contains a table with two columns: 'Domain' and 'Status'. It lists three domains: 'acmebank.com' (Status: Active), 'acmebank.org' (Status: Inactive), and 'acmebank.net' (Status: Active). The right card is titled 'Monitoring' and contains a table with two columns: 'Metric' and 'Value'. It lists three metrics: 'Uptime' (Value: 99.99%), 'Latency' (Value: 10ms), and 'Error Rate' (Value: 0.01%).

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login:
acmebank.com/login
logo hash: 3f9a...
screenshot hash: a21c...

Regions: US, EU
Score threshold: 0.65

Per-brand config: canonical assets, regions, escalation threshold.

P05

Triage queue

app.spoofvane.io

SpoofVane

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Triage queue 247

Verdict	URL	Score	Source	SLA
PHISH	acme-secure-login.top	0.91	SERP	2m
SUSPICIOUS	acmebank-verify.xyz	0.58	OpenPhish	14m
PHISH	my-acme.cc	0.87	Registrar	5m
BENIGN	acme-fanblog.net	0.12	CertStream	1h

Tenants

Settings

Analyst's main workspace. Click a row → Alert detail.

P06 Alert detail — 10 evidence tabs

app.spoofvane.io

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Settings

acme-secure-login.top PHISH

Request takedown

SummaryEvidenceMulti-regionVerdict traceMITREKitClusterDeepfakeTakedownAudit

Screenshot (US-residential)

rendered via Bright Data Scraping Browser

Verdict trace

claude: phish 0.92
gpt-5: phish 0.89
gemini: phish 0.90
merge: PHISH · no dissent

The heart of the product. Ten tabs of evidence; takedown request starts here → Review queue.

P07 Clusters

app.spoofvane.io

SpoofVane

Dashboard

Threat-network graph

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Settings

acme-secure.top

– ASN AS200651 –

my-acme.cc

– kit EvilProxy –

acme-verify.xyz

Nodes link by shared ASN / registrar / kit / favicon / JA3.

Reveals campaigns, not single sites. Click a node → its alert.

P08 Deepfakes

app.spoofvane.io

SpoofVane

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Deepfake review

CEO impersonation clip #1182 **PHISH**
face dist 0.81 · lip-sync 0.74 · voiceprint match 0.03 · C2PA
x invalid
multimodal verdict: DEEPFAKE 0.79

Users
Tenants
Settings

Multimodal queue: face, lip-sync, voiceprint, C2PA provenance.

P09 Exec protection

app.spoofvane.io

 SpoofVane

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Executive protection lawful-basis gated

Jane Doe, CEO

Likeness assets found: 3 · Voice assets: 1

Impersonation handles: @jane_x_15,
@jane_linkedin_57

Sensitive surface — needs a recorded lawful basis before any action.

P10 Takedowns

app.spoofvane.io

 SpoofVane

Takedown status

Dashboard	URL	Channel	Status	Ref
Brands	acme-secure.top	Cloudflare	submitted	CF-6d8f
Triage	my-acme.cc	Hetzner abuse	HITL pending	–
Clusters	acme-verify.xyz	Namecheap	acknowledged	NC-44a1
Deepfakes				
Exec				
Takedowns				
Audit				
Review				
Cost				
Compliance				
Agents				
Users				
Tenants				
Settings				

Registrar + hosting-abuse channels, reference IDs, manual override.

P11 Audit

app.spoofvane.io

SpoofVane

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Audit log

Ask: analyst takedown actions for tenant acme

#	Actor	Action	Outcome	Hash
1182	analyst:kim	takedown_request	ok	a3f9...
1183	reviewer:lee	takedown_approve	ok	b2c1...

Agents
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Settings

Hash-chained, NL-searchable (H2). Tamper-evident.

P12 Review queue

app.spoofvane.io

 SpoofVane

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HITL review

reviewer $\neq$ analyst

Takedown: acme-secure.top

PHISH

Raised by analyst:kim · awaiting reviewer approval

Approve

Deny

Consequential actions wait here for a different person's approval.

P13 Cost

app.spoofvane.io

 SpoofVane

Bright Data spend

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Settings

Tenant: acme · tier: business

\$412 / \$1000 envelope · SLM cheap-path 73%

Spend vs tier envelope, budget alerts, SLM-first routing.

P14

Compliance

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Compliance evidence

SOC 2

ISO 27001

DORA

NIS2

Control status + exportable auditor pack.

Agents
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Settings

What auditors read: framework status + evidence export.

P15 Admin · Agents

app.spoofvane.io

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Agents

× Global kill-switch

Agent	State	Blast
takedown_agent	idle	0.6
cred_poison_agent	blocked (2-person)	0.85

Registry, governance budgets, and the kill-switch (E7).

P16 Admin · Users

app.spoofvane.io

SpoofVane

Users & roles

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User	Role
kim	analyst
lee	reviewer

RBAC — 6 roles with segregation of duties.

P17

Admin · Tenants

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Tenants

Tenant	Residency	Tier
acme	EU	business

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Multi-tenant management + data-residency region per tenant.

P18

Admin · Demo health



app.spoofvane.io

 SpoofVane

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Settings

Demo health

Pages covered: 21/21 ✓

Seed brands: 3 ✓

BD products exercised: 7/7 ✓

Internal — seed completeness + 21-page coverage check.

P19

Settings



app.spoofvane.io

 SpoofVane

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Settings

Profile · MFA · API keys

MFA: enabled  · API keys: 2 active

Profile, MFA enrolment, API keys.

P20 Not found (404)



app.spoofvane.io

404

This surface does not exist.

Shown for unknown URLs.

P21 Forbidden (403)



app.spoofvane.io

403

Your role does not permit access.

| Shown when your role lacks access.

SpoofVane Visual User Guide – screens mirror the canonical page registry
console/src/lib/pages.ts. Mockups use the product's real design tokens
(console/src/tokens/tokens.css).